Department of Computer Science and Engineering
University of Barisal

Course: Theory of Computation Marks: 50, Time: 50 Min

Instructor: Dr Md Manjur Ahmed

Question:

1) What is Automata Theory? Describe the applications of Finite Automata. (6 Marks)
2) Explain the work of lexical analyzer with an on/off switch. (3 Marks)
3) Analyze the terms: Theorems, Lemmas and Corollaries. (6 Marks)
4) Construct a NFA for the following: Strings where the first symbol is present somewhere later on at least once. (5 Marks)
5) Build an NFA for the following language: L = { w | w ends in 101} (10 Marks)
6) Convert the NFA from Question 5, to DFA. (5 Marks)
7) Build an ε-NFA for the following language:
   L = {w | w is empty, or if non-empty will end in 11} (5 Marks)
8) Convert ε-NFA to DFA based on Question 7. (10 Marks)

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